



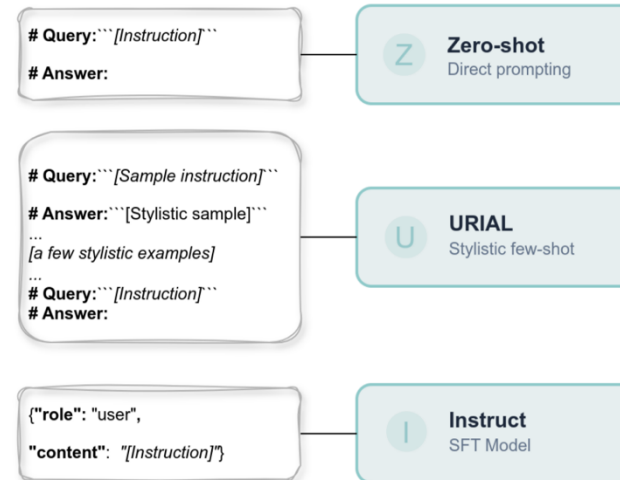
University
of Exeter

COMM117

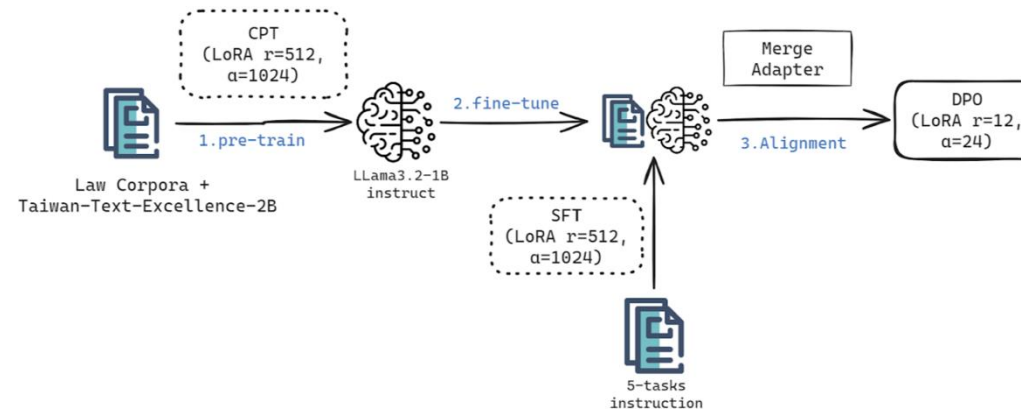
Large Language Models and Application

Two families of approaches

Prompting / In-context learning



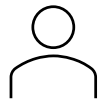
Finetuning (SFT)



[Efficient Training of Robust Traditional Chinese LLaMA-1B on a Single Consumer GPU: Continual Pre-training, SFT, and DPO](#)

Baseline: Zero-shot prompt

- Zero-shot translation = “Translate” instruction + input text, with **no in-prompt examples (and no fine-tuning)**.
- You just give an instruction like “Translate X to Y” and the model relies on its general pretrained multilingual ability.

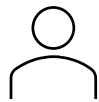


Translate the following English text into French:
“The system will be unavailable from 2:00 to 3:30 PM for scheduled maintenance.”



« Le système sera indisponible de 14 h 00 à 15 h 30 en raison d’une maintenance planifiée. »

“zero-shot with constraints” (still zero-shot)



Translate into Spanish.

Constraints: keep numbers and times exactly, keep product name “DataHub” unchanged.

Text: “DataHub will be unavailable from 2:00 to 3:30 PM.”

Baseline: Zero-shot prompt

- **Simplest prompt**

You give the most basic instruction, e.g. no examples, no glossary, no extra rules.

- **Surprisingly strong**

Modern LLMs often produce **fluent, good** instruction—especially for common language tasks.

- **Fragile on constraints**

The same minimal setup often **fails with**

- keep **numbers/dates** exactly unchanged
- preserve **HTML/XML tags** exactly

HTML tag break

extra/missing closing tag,
tag moved,
or punctuation inserted inside tags.

ChatGPT 5.2 ▾

Translate into chinese. Preserve the HTML exactly (all tags must remain exactly the same and correctly closed). Text: Click **Save** to continue. If you see **<i>Error 17</i>**, contact support.

点击 **保存** 以继续。如果您看到 **<i>Error 17</i>**，请联系支持。

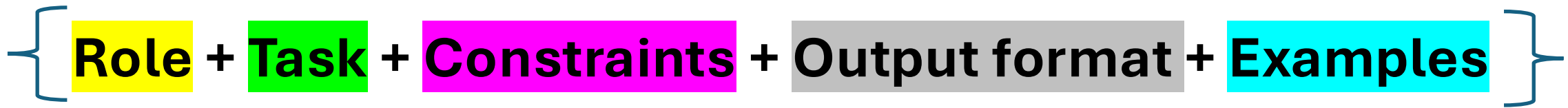


Translate into Spanish. Preserve the HTML exactly (all tags must remain exactly the same and correctly closed).

Text:
Click **Save** to continue. If you see **<i>Error 17</i>**, contact support.

Haga clic en **Guardar** para continuar. Si ve **<i>Error 17</i>**, contacte con el soporte técnico.

Prompting pattern



Prompting pattern 1: “constraints-first” translation

Prompting pattern 2: Few-shot with style + glossary

Prompting pattern 3: Structured output (JSON / XML)

Prompting pattern 4: Two-pass translate → revise

Prompting pattern 5: Self-check (targeted)

- List changed entities
- List changed numbers
- Glossary compliance: yes/no



Demo: Fine-tuning an MT model via SFT

Setup

- ✓ Model: Helsinki-NLP/opus-mt-en-de (small, fast)
- ✓ Method: LoRA (parameter-efficient)
- ✓ Packages: transformers, datasets, evaluate, peft, accelerate

```
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

model_name = "Helsinki-NLP/opus-mt-en-de"
tokenizer = AutoTokenizer.from_pretrained(model_name)
base_model = AutoModelForSeq2SeqLM.from_pretrained(model_name)
```

Dataset

- 60–200 examples is enough to demo behavior change
- Include “hard cases”:
 - {placeholders} unchanged
 - UI terms fixed (“Account”→“Konto”)
 - HTML tags preserved
- Split: train / validation / test (e.g., 80/10/10)

- {"src":"Hi {first_name} ...","tgt":"Hallo {first_name} ...","type":"placeholder"}
- {"src":"Click Save","tgt":"Klicken Sie auf Save","type":"html"}

Preprocessing (tokenization + labels)

- Tokenize **src** as encoder **input**
- Tokenize **tgt** as decoder **labels**

```
model_inputs = tokenizer(
    batch["src"], max_length=max_source_len, truncation=True
)
with tokenizer.as_target_tokenizer():
    labels = tokenizer(
        batch["tgt"], max_length=max_target_len, truncation=True
    )["input_ids"]
```

Evaluation

- **Automatic**

BLEU score(word n-gram overlap); *chrf* (character n-grams)

- **Human rubric**

Adequacy (meaning preserved)

Terminology/Consistency

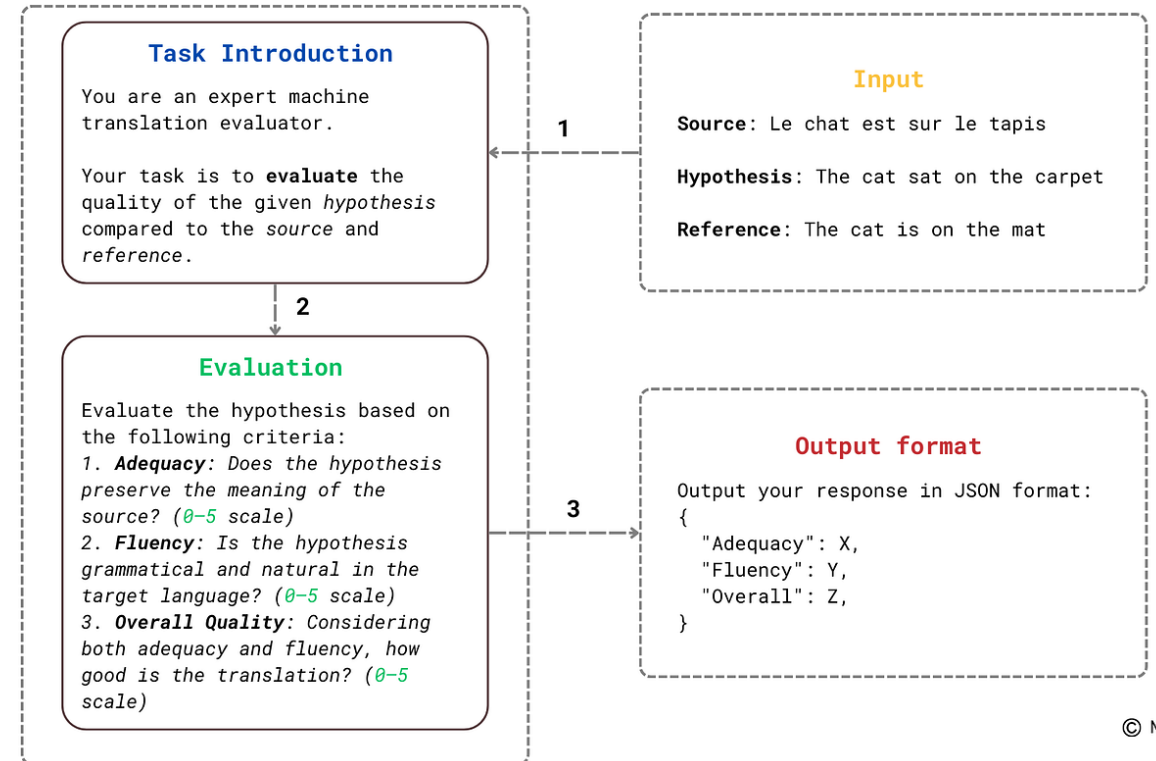
Format/Constraints:

- **Constraint checks**

Exact –match check (place holder/tags...)

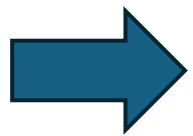
Glossary check

Length / structure check



Take Away

- Use prompting for speed/experimentation
- Fine-tune for consistency at scale
- Always evaluate constraints explicitly



Translation often needs **external knowledge**: terminology
DBs, product names, policies → retrieval

RAG

Introduction Today

- RAG: Retrieval-Augmented Generation
 - Why
 - How
 - Evaluation

RAG: Retrieval-Augmented Generation

LLMs are fluent → But not always faithful → External knowledge fixes many failures

- LLMs **hallucinate**
- Knowledge is **frozen at training time**
- Enterprises have **private data**
- **Prompts** alone are **not enough**

ChatGPT 5.2 Instant ▾

Give me a **verbatim** quote (2–3 sentences) about RAG evaluation

From the paper *"Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks"* by researchers at Facebook AI:

"We introduce Retrieval-Augmented Generation (RAG) models which combine pre-trained parametric and non-parametric memory for language generation. RAG models access a dense vector index of Wikipedia, enabling them to produce more factual and specific language than parametric-only seq2seq models. We evaluate RAG on open-domain question answering and achieve state-of-the-art results on Natural Questions, WebQuestions, and CuratedTrec."

Verbatim

Not verbatim!

RAG: Retrieval-Augmented Generation

LLMs are fluent → But not always faithful → External knowledge fixes many failures

- LLMs **hallucinate**
- Knowledge is **frozen at training time**
- Enterprises have **private data**
- **Prompts** alone are **not enough**

Latest Research about RAG

latest research about RAG

Here's a focused, up-to-date research overview of Retrieval-Augmented Generation (RAG) in 2026, covering recent papers, active themes, and notable developments in the field.

1) Very Recent Academic Research (2025–2026)

1.1 IGMiRAG: Intuition-Guided RAG (2026)

Title: IGMiRAG: Intuition-Guided Retrieval-Augmented Generation with Adaptive Mining of In-Depth Memory

✗ 2026.2.7

TCM-DiffRAG: Personalized Syndrome Differentiation Reasoning Method for Traditional Chinese Medicine based on Knowledge Graph and Chain of Thought

J Li, Y Chang, SK Tang, Y Liu, Y Wang, S Lin... - arXiv preprint arXiv ..., 2026 - arxiv.org

3 days ago - ... Improving large language model applications in biomedicine with **retrieval-augmented generation**: a systematic review, meta-analysis, and clinical development guidelines...

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📄 Related articles

🔗

2026.2.26

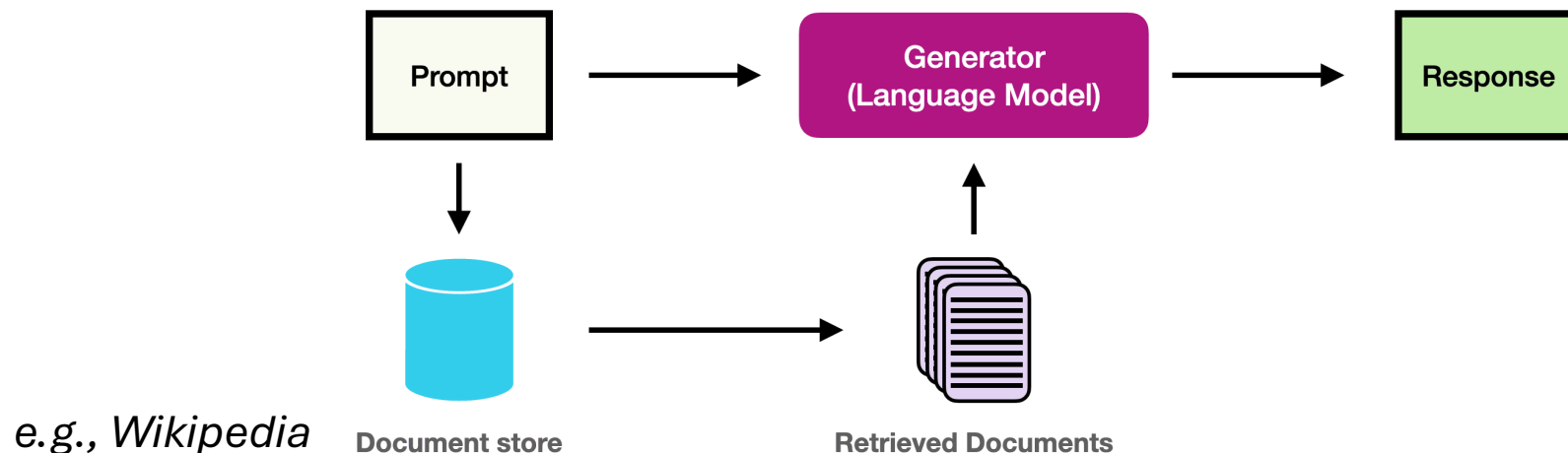
Fine-Tuning not enough

- **Forgetting/ losing capabilities :**
e.g. a finance fine-tuned LLM can't handle general conversational tasks well.
- **Training data dependence:**
Performance is entirely dependent on the **quantity and quality** of available training data. Collecting high-quality data is expensive.
- **Lacks external knowledge:**
The model only knows what's in its training data, and lacks **real-world knowledge**.
- **Not customizable:**
 - Changes to the fine-tuned model require **retraining which is expensive**.

RAG: Retrieval-Augmented Generation

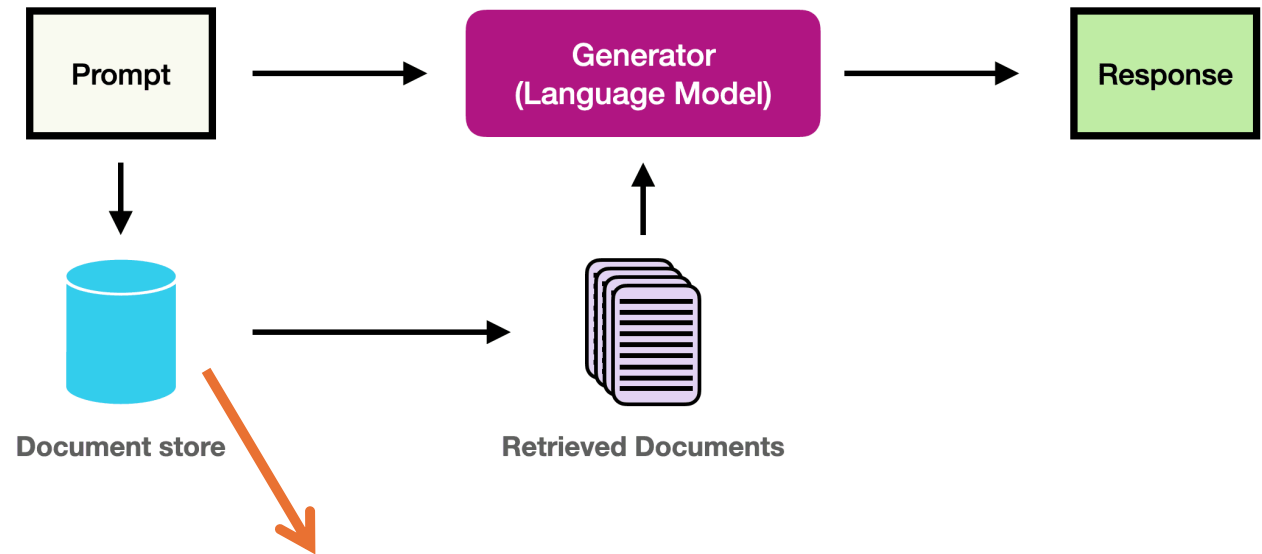
“We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models **which combine pre-trained parametric** and **non-parametric memory for language generation**. We introduce RAG models where the parametric memory is a **pre-trained seq2seq model** and the non-parametric memory is **a dense vector index of Wikipedia, accessed with a pre-trained neural retriever**.”

-- [*Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*](#) , Patrick Lewis (Facebook AI Research) & Ethan Perez



RAG: Retrieval-Augmented Generation

“We explore a general-purpose fine-tuned (RAG) — models **which combine** pre-trained seq2seq model and the Wikipedia, accessed with a pre-trained model — *Retrieval-Augmented Generation for Knowledge* — Ethan Perez



Parametric

Inside weights

Hard to update

Expensive retraining

Non-parametric

External DB

Easy to update

Cheap indexing

RAG: Retrieval-Augmented Generation

RAG Can

- **Access to fresh information**

Pretrained data is limited → RAG can provide **up-to-date information**

- **Factual grounding**

Providing “facts” to the LLM as part of the input prompt can **mitigate hallucinations**

- **Long documents input**

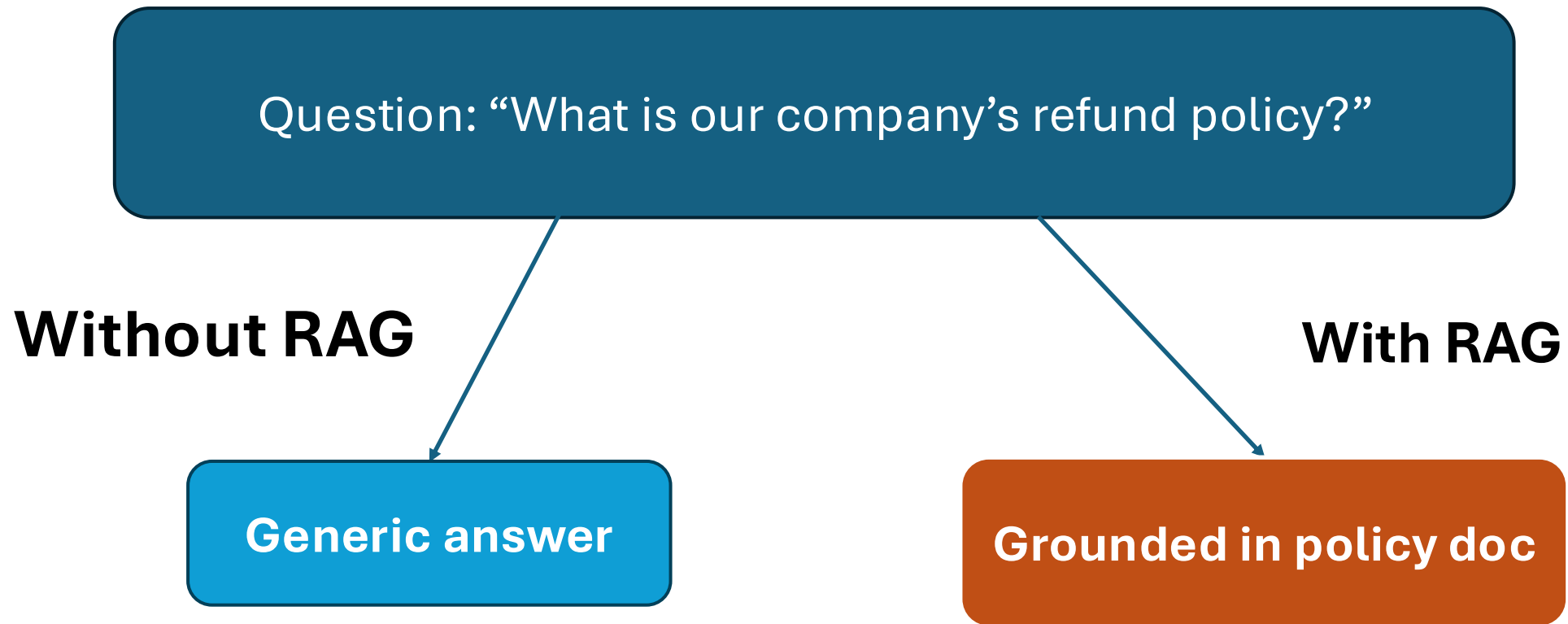
You can't fit (or reliably attend to) a 200-page manual in the prompt

- **Evidence-backed outputs**

Show citations + quoted spans for critical facts

RAG: Retrieval-Augmented Generation

RAG Can

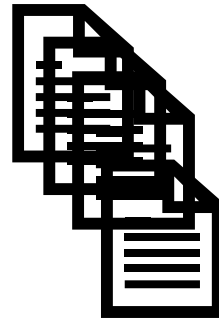


We reduce hallucination by forcing evidence.

What is Retrieval?

Retrieval is the process of **accessing stored information** from memory and **bringing it into conscious awareness.**

- **Find relevant documents**
- **Ranking problem**
- **Similarity search**



Which of these 10,000 chunks are most relevant according to your query?

- Ranking
- Similarity search

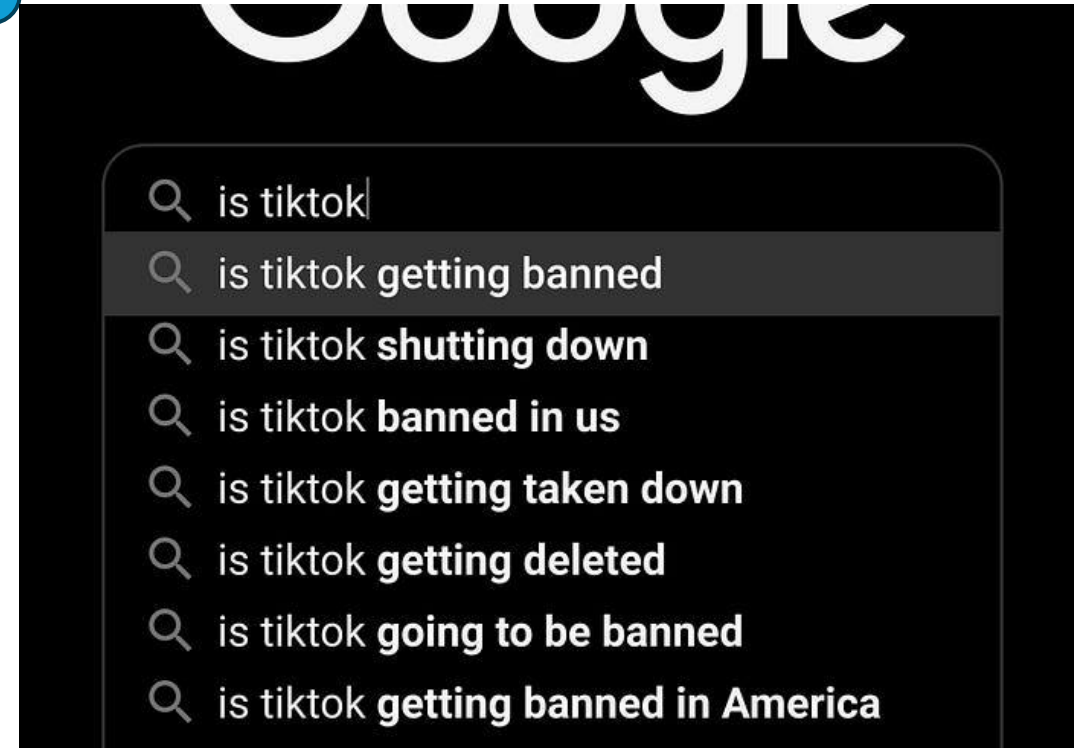
What is Retrieval?

Lexical Retrieval (TF-IDF / BM25)

A **search method** that matches documents to a query based on **exact words (tokens)** and their frequency.

- It does **not** understand meaning.
- It works at the **word level**, not semantic level.

- Find relevant documents



What is Retrieval?

Lexical Retrieval (TF-IDF / BM25)

$$TF(t, d) = \frac{(\text{Number of occurrences of term } t \text{ in document } d)}{(\text{Total number of terms in the document } d)}$$

$$IDF(t, D) = \log_e \frac{(\text{Total number of documents in the corpus})}{(\text{Number of documents with term } t \text{ in them})}$$

$$TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

FINAL RELEVANCE SCORE:
The estimated relevance of document D to query Q.

TERM FREQUENCY (TF)
Number of times query term q_i appears in document D.

k_1 PARAMETER
Controls term frequency saturation. Higher k_1 delays saturation (rewards frequent terms more linearly). Typical values: 1.2 – 2.0.

DOCUMENT LENGTH
Total number of terms in document D.

$$Score(D, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \left(\frac{|D|}{avgdl}\right)\right)}$$

INVERSE DOCUMENT FREQUENCY (IDF)
Measures term rarity across the entire collection. Penalizes common terms, boosts rare, informative terms.

b PARAMETER
Controls the degree of length normalization

DOCUMENT LENGTH NORMALIZATION
Adjusts for document

AVERAGE DOCUMENT LENGTH

- **Term frequency** (how often word appears)
- **Inverse document frequency** (rare words matter more)
- **Document length normalization**

What is Retrieval?

Lexical Retrieval (TF-IDF / **BM25**)

Word/tokens

Dense Retrieval (Embeddings)

Embeddings/ **Vectors**

“Dense Retrieval is a technique that transforms **queries and documents** into **continuous, low-dimensional embeddings**, enhancing **semantic similarity** beyond simple lexical matching.”

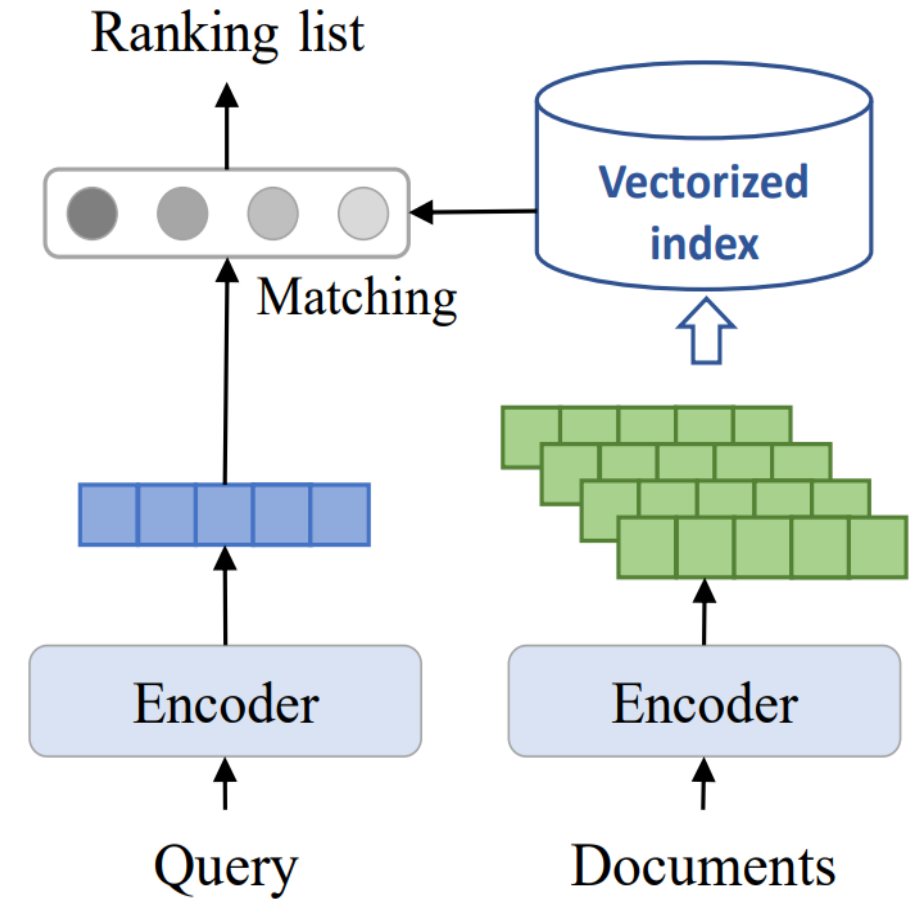
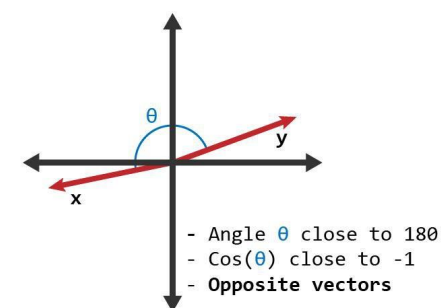
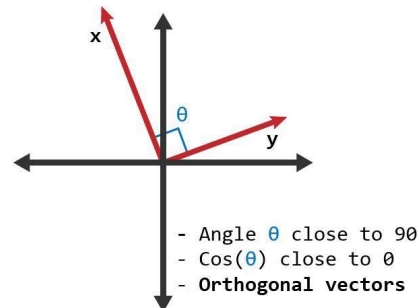
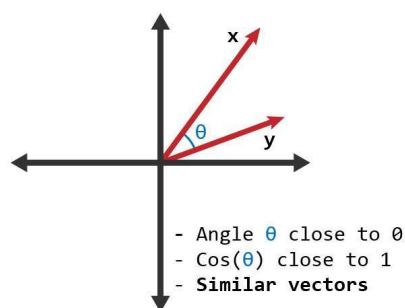
What is Retrieval?

Dense Retrieval (Embeddings)

Why embeddings?

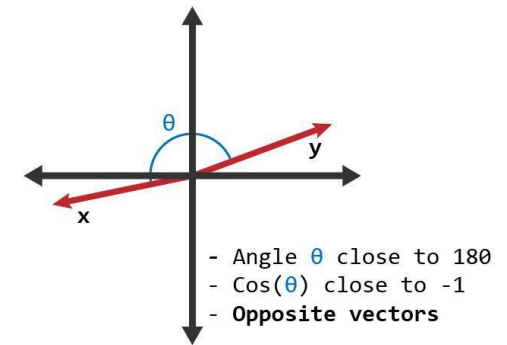
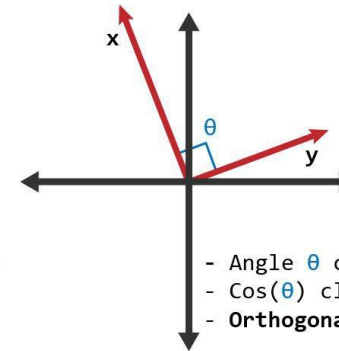
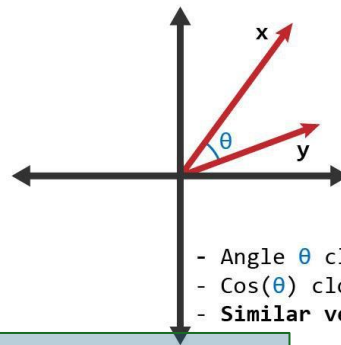
Semantic similarity, not keyword overlap.

Cosine similarity



Dense Retrieval

Cosine Similarity



Document 1 = 'the best data science course'

Document 2 = 'data science is popular'

V	the	best	data	science	course	is	popular
D1	1	1	1	1	1	0	0
D2	0	0	1	1	0	1	1

$D1 = [1, 1, 1, 1, 1, 0, 0]$

$D2 = [0, 0, 1, 1, 0, 1, 1]$

Dot product $D1 \cdot D2 = 1 \times 0 + 1 \times 0 + 1 \times 1 + 1 \times 1 + 1 \times 0 + 0 \times 1 + 0 \times 1 = 2$

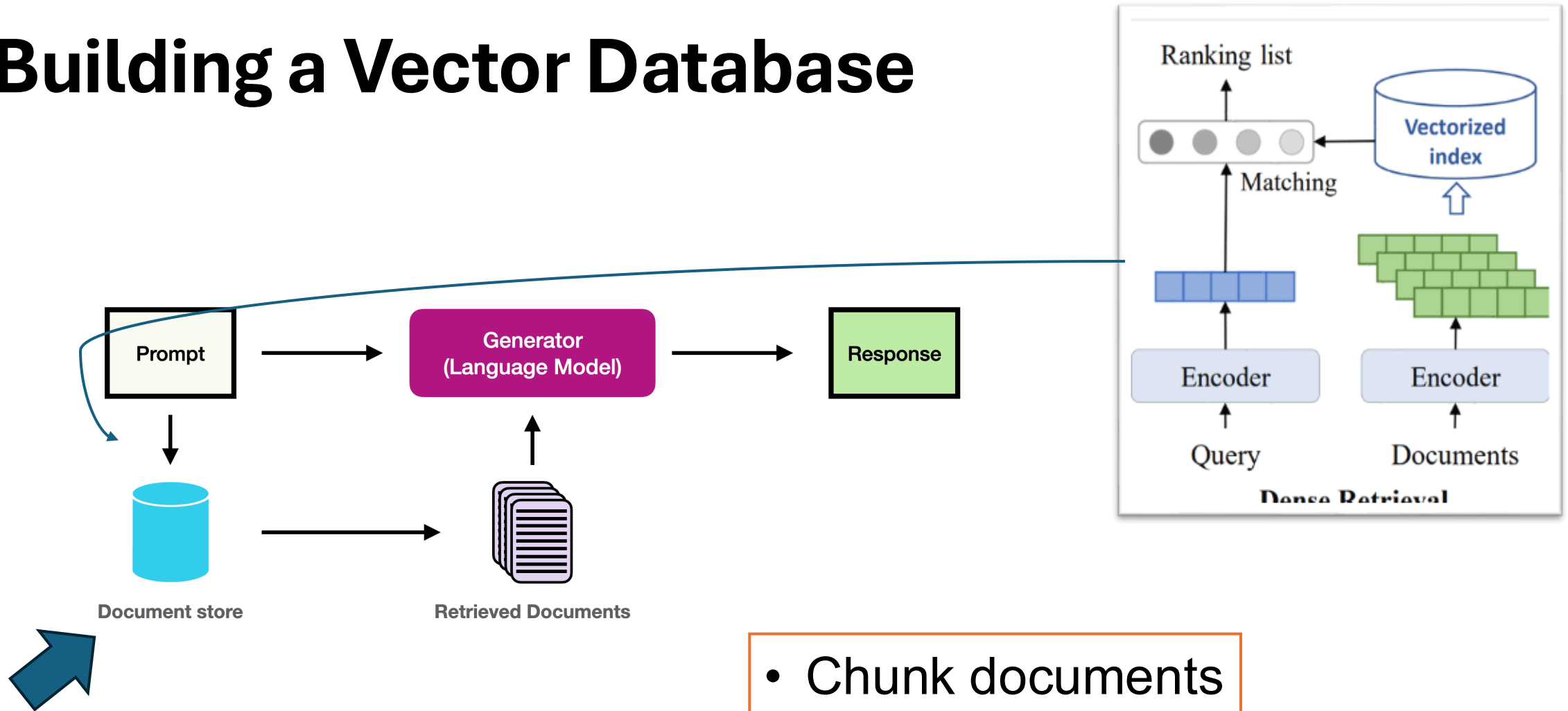
Magnitude

$$\|D1\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 0^2 + 0^2} = \sqrt{5} \quad \|D2\| = \sqrt{0^2 + 0^2 + 1^2 + 1^2 + 0^2 + 1^2 + 1^2} = \sqrt{4}$$

$$\text{similarity}(D1, D2) = \frac{D1 \cdot D2}{\|D1\| \|D2\|} = \frac{2}{\sqrt{5}\sqrt{4}} = \frac{2}{\sqrt{20}} = 0.44721$$

$$\theta = 63.345$$

Building a Vector Database



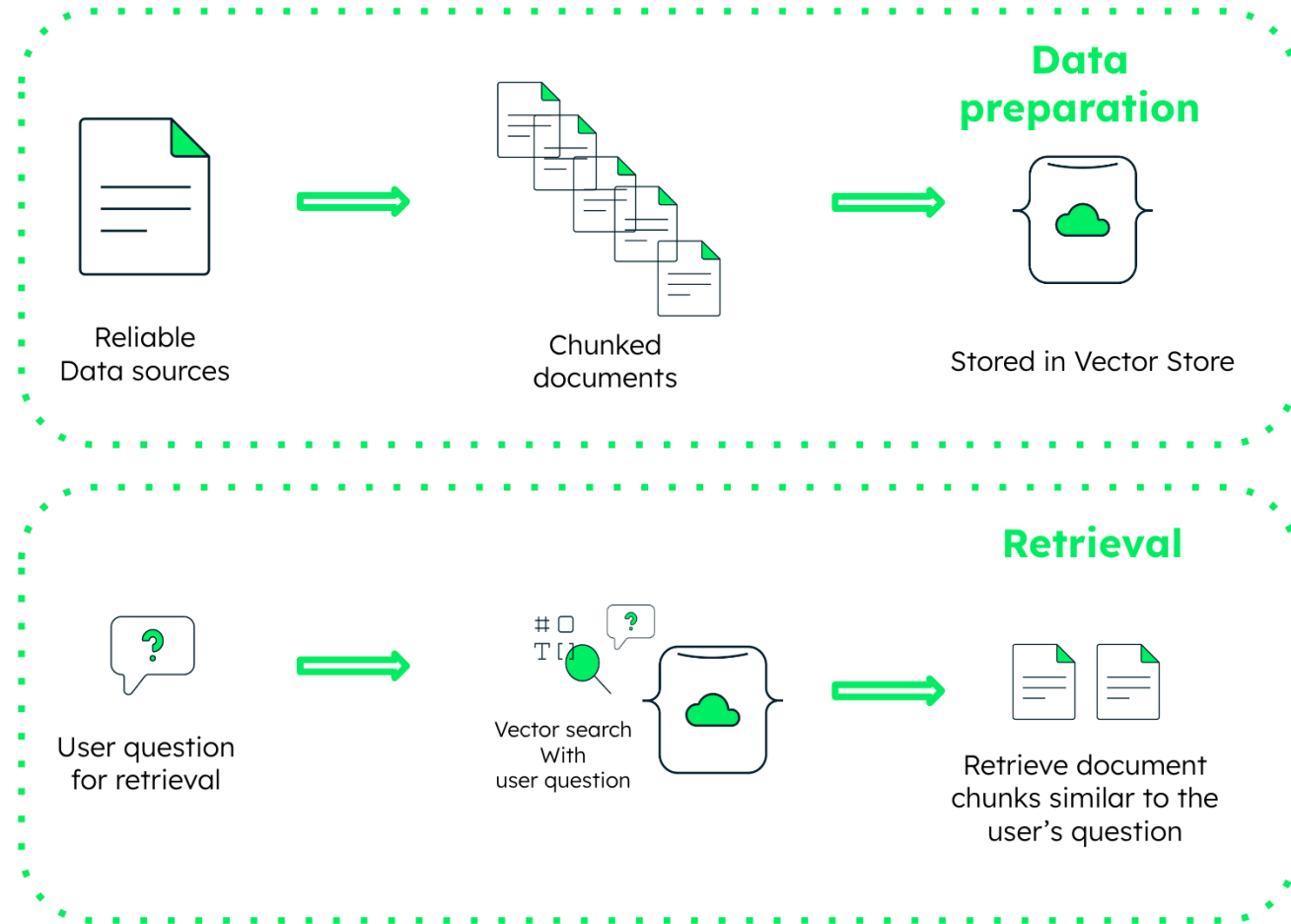
- Chunk documents
- Embed chunks
- Store vectors
- Search top-k

Chunking

Chunking is the process of splitting large documents into **smaller pieces (chunks)** before embedding and storing them in a vector database.

Instead of embedding an entire 20-page PDF, you break it into smaller segments like:

- Paragraphs
- Sections
- Sliding windows of 500 tokens
- Semantically meaningful units



Chunking- Why

1. Embeddings work better on **focused text**

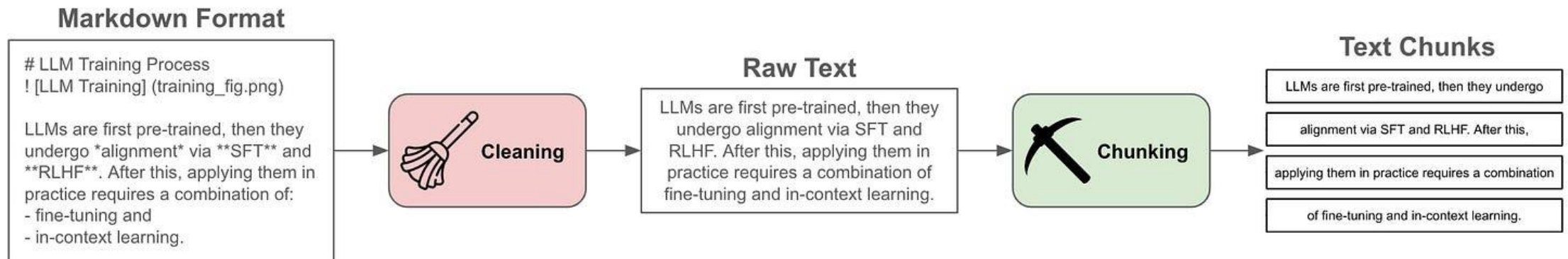
Embedding an entire 10-page document produces a very “averaged” vector.

2. LLM **context window is limited**

You cannot insert entire books into the prompt.

3. Retrieval precision

You want to **retrieve only the part** that answers the question, not the whole document.



Chunking

- **Fixed-size chunking**
Split every 500 **tokens**. → Simple but may cut mid-sentence.
- **Sentence-based chunking**
Group **sentences** until token limit reached. → More natural.
- **Section-based chunking**
Use document **headings**. → Best when structure exists.
- **Semantic chunking (advanced)**
Split where **topic changes** using embedding similarity.
More complex but high quality.
- **Agentic Splitting:**
Employing **LLM** to determine optimal chunking based on content and context.

“Chunking quality often matters more than model choice.”

Chunking – Fixed-size chunking



© Image created by Han HELOIR

Most common

Chunking – fixed-size

Good Chunking

Usually:

- 300–800 tokens
- Sometimes 10–20% overlap
- Logically coherent (same topic)

Chunk 1: tokens 0–500

Chunk 2: tokens 400–900

Why overlap?

To avoid cutting important information mid-thought.

Without overlap:

Important sentences might split across chunks.

Chunking – Semantic chunking- advanced



The Olympic Games began in ancient Greece, held every four years in Olympia from 776 BC to 393 AD. They were a central part of Greek culture, bringing together athletes from various city-states. The Games were revived in 1896 by Pierre de Coubertin, aiming to create a modern Olympics. The first modern Games were in Athens, Greece, with fourteen nations participating. Today, the Olympics are the world's foremost sports competition, featuring athletes from nearly every country and alternating every two years between the Summer and Winter Games.



Step 1: Sentence Splitting

1. "The Olympic Games began in ancient Greece, ..."
2. "They were held every four years in Olympia ..."
3. "They were a central part of Greek culture, ..."
4. "The Games were revived in 1896 by Pierre ..."
5. "The first modern Olympic Games were held in ..."
6. "Today, the Olympics are the world's foremost ..."



Step 2: Generate Embeddings

- **Generate embeddings** for each sentence
- Capture semantic meaning in vector form



Step 3: Group Sentences by Similarity

- **Compare embeddings** to find semantically similar sentences (Cosine Similarity)
- **Group sentences** into chunks based on thematic (Thresholding or Clustering)

Chunking

Semantic Chunking Example

Group two sentence together into Chunk 1:

Explanation: Similar meaning of two sentences on the origins & Cultural Significance of Olympic
Calculated cosine similarity of sentence before and after is 0.95

Chunk Threshold: 0.8

The Olympic Games began in ancient Greece, held every four years in Olympia from 776 BC to 393 AD. They were a central part of Greek culture, bringing together athletes from various city-states. The Games were revived in 1896 by Pierre de Coubertin, aiming to create a modern Olympics. The first modern Games were in Athens, Greece, with fourteen nations participating. Today, the Olympics are the world's foremost sports competition, featuring athletes from nearly every country and alternating every two years between the Summer and Winter Games.

Chunk point (Do not group):

Explanation: Sentence before: Olympic history, sentence after; Olympic revival
Calculated cosine similarity of sentence before and after is 0.3

Chunk 3 :

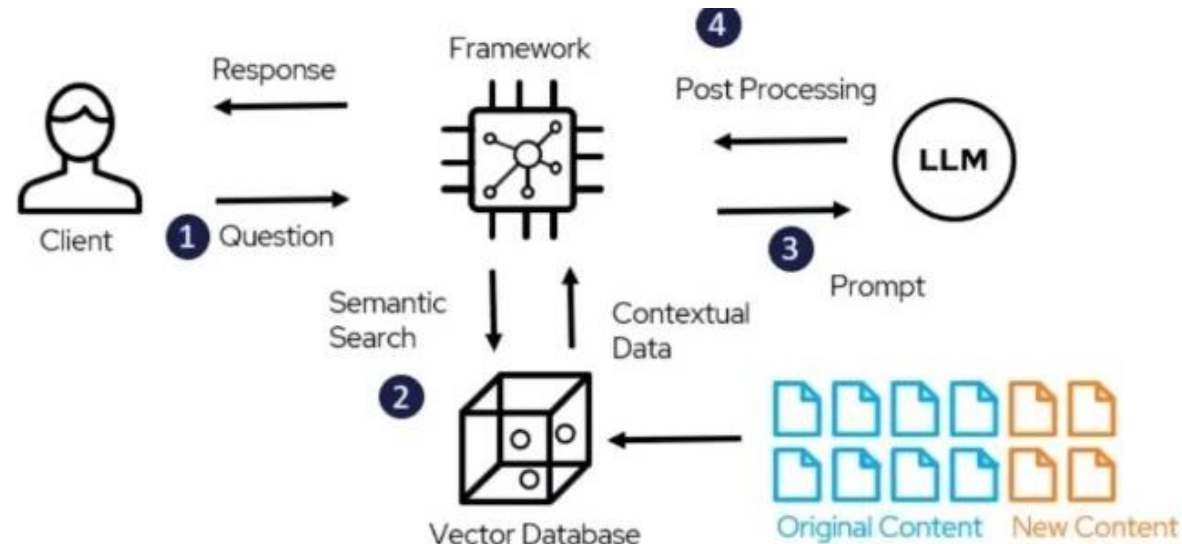
Explanation: Not similar with the sentence before.
Calculated cosine similarity of sentence before and after is 0.4

Group two sentence together into Chunk 2:

Explanation: Similar meaning of two phrases on the Revival & Modernization of Olympic
Calculated cosine similarity of sentence before and after is 0.82

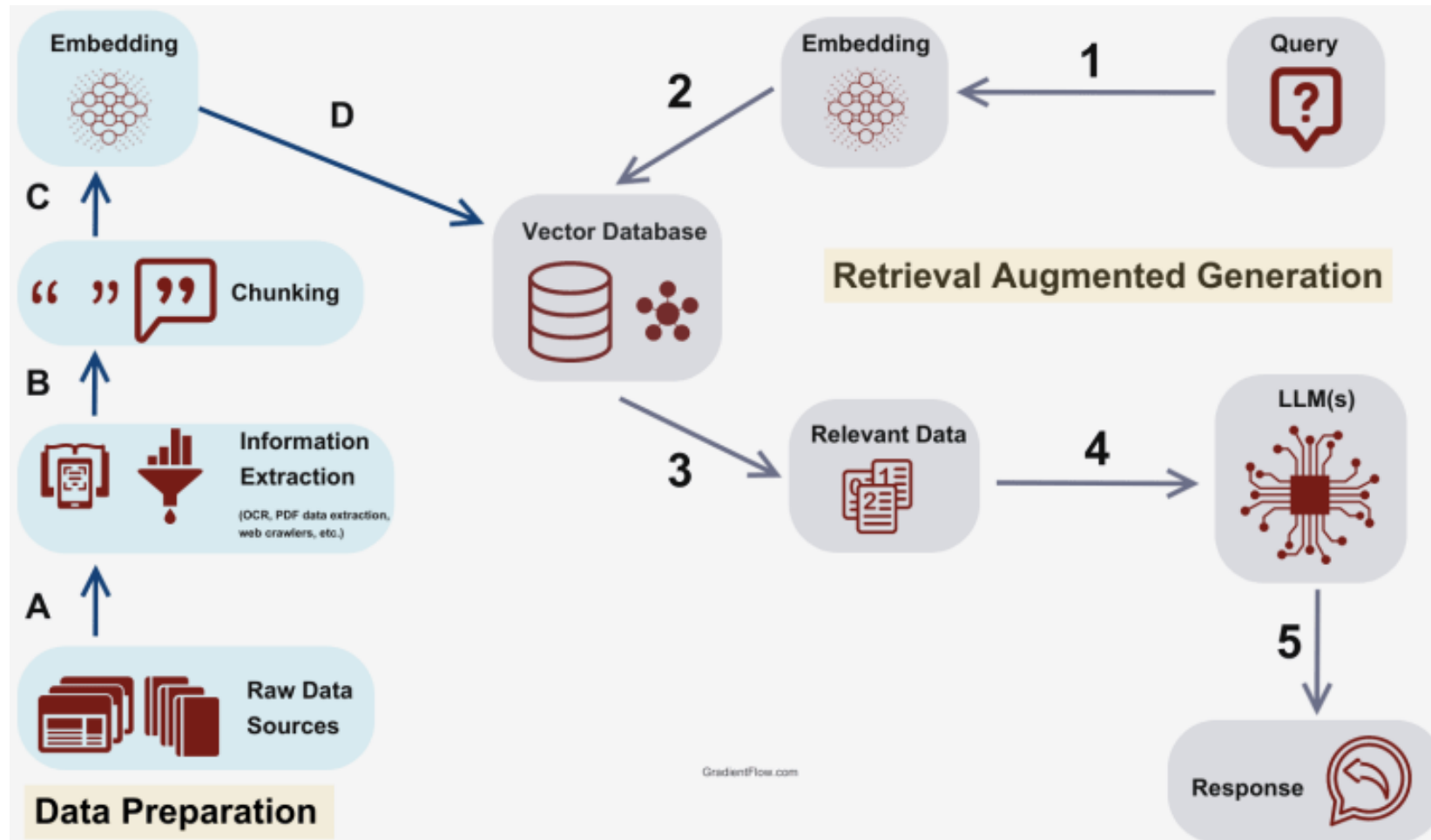
RAG

*“A RAG process takes **a query** and **assesses** if it relates to subjects defined in the **paired knowledge base**. If yes, it searches its knowledge base to **extract information** related to the user’s question. Any **relevant context in the knowledge base** is then passed to the **LLM** along **with the original query**, and an answer is produced.”*



RAG- pipeline

Query → Embed → Retrieve → Prompt Template → LLM → Answer



RAG- Prompt Template

Example:

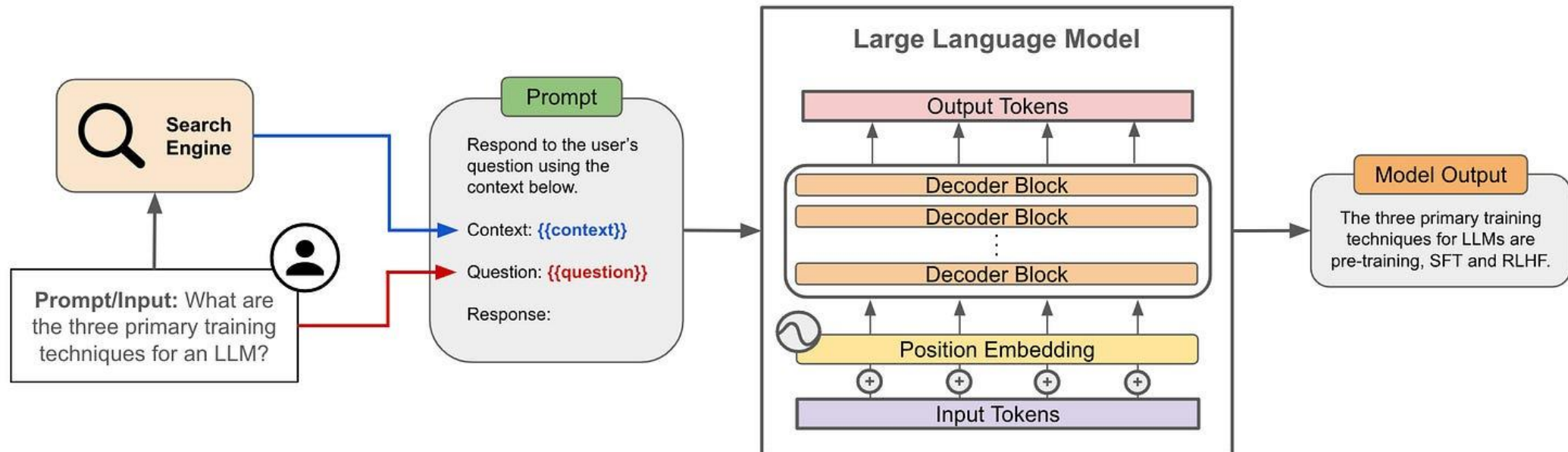
You are a QA system. Use ONLY the context below. If unsure, say “Not found.”

Context:

{retrieved_chunks}

Question:

{query}

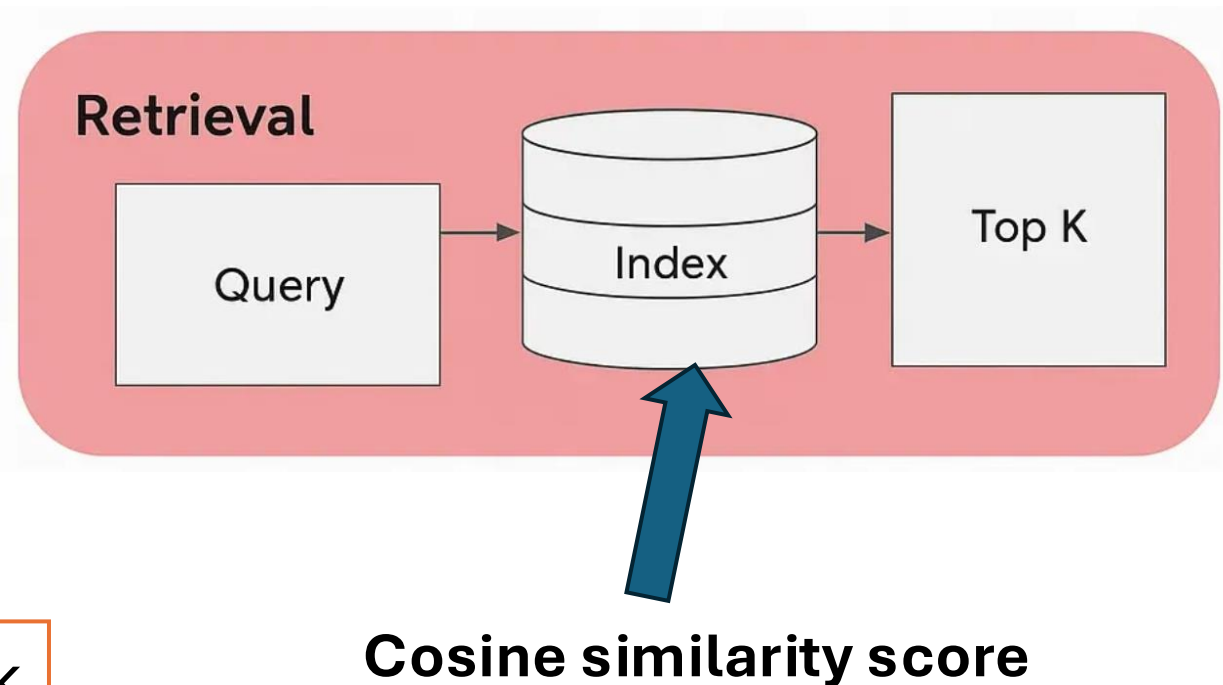


Retrieval

Retrieval Quality = RAG Quality

When a user submits a query, the system compares the query's embedding against the indexed embeddings and retrieves the **top-k most similar chunks**.

If relevant document $\notin$ Top-k
→ Answer cannot be correct



Reranking

- k too small → missing info
- k too large → context overflow

Hard to decide!

Sometimes →

- The most relevant document is ranked 3rd or 5th.
- Slightly noisy chunks appear above better ones.



Two-Stage Retrieval Architecture

Stage 1: Fast Retriever

Select Top-50 candidate chunks



Stage 2: Cross-Encoder Re-ranker

Reordered Top-5



LLM

Reranking

Stage 1: Fast Retriever

Select Top-50 candidate chunks

User query:

“Can I refund a digital product?”

Retriever returns Top-50 candidates:

Chunk 17 – Refund policy section

Chunk 83 – Digital goods clause

Chunk 12 – Shipping policy

Chunk 204 – Account FAQ

...

(50 chunks total)

Stage 2: **Cross-Encoder Re-ranker**

Reordered Top-5

LLM

A **cross-encoder re-ranker** is usually:

A **Transformer encoder model (often BERT-style)** fine-tuned for relevance scoring. → Encoder only Transformer

A classification/scoring model

- BERT
- RoBERTa
- DeBERTa
- MiniLM



scores



Ranking Top-5 selected

Retrieval- Hybrid search

Hybrid retrieval means:

Combining lexical search (e.g., BM25) and dense retrieval (embeddings) together.

Because:

- Lexical search is good at exact keyword matching.
- Dense search is good at semantic similarity.

Each has weaknesses.

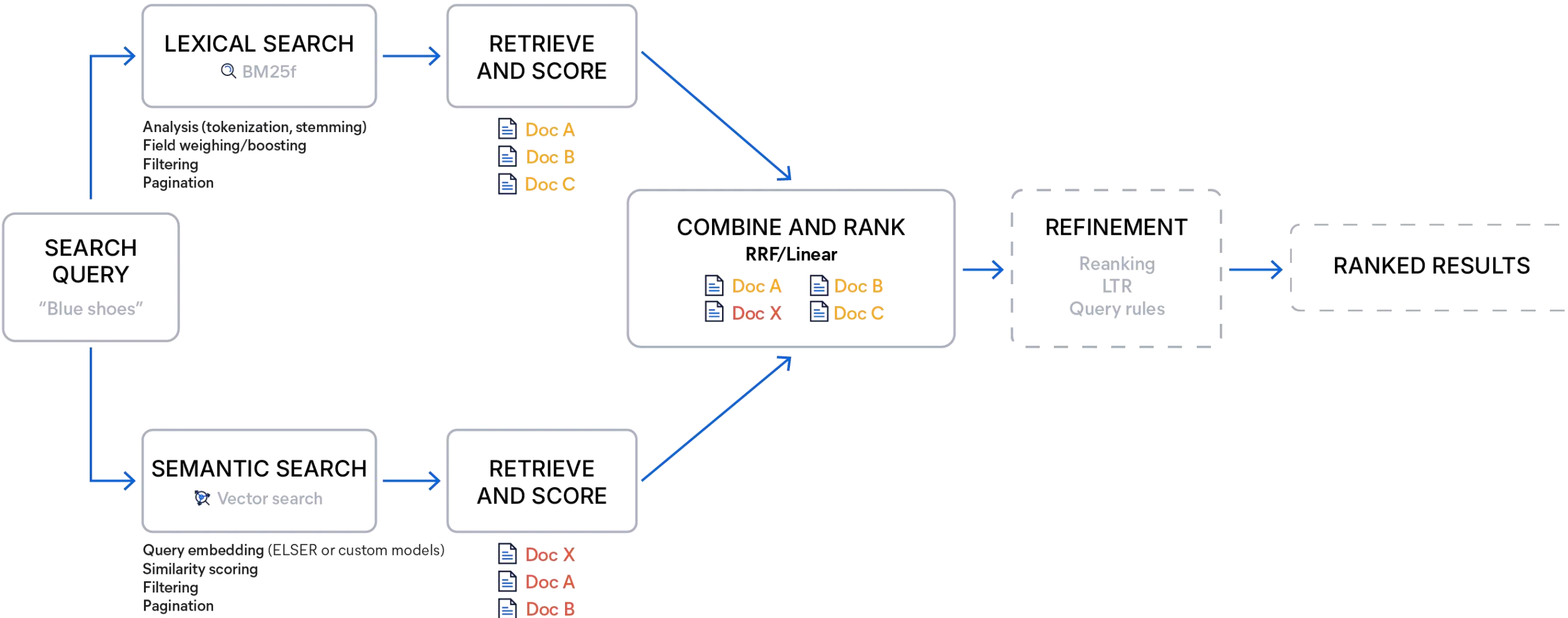
Lexical Retrieval (TF-IDF / BM25)

Word/tokens

Dense Retrieval (Embeddings)

Embeddings/ **Vectors**

Retrieval- Hybrid search



Retrieval- Hybrid search

Query:

Error code 0xA93 refund failure

Document A:

Refund system failure due to payment timeout.

Document B:

Error code **0xA93** occurs when authentication fails.

**Dense Retrieval
(embeddings)**

Rank **A** high → semantic match

Lexical retrieval (BM25)

Rank **B** high → exact match:
0xA93

Retrieval- Hybrid search

Query:

Error code 0xA93 refund failure

Document A:

Refund system failure due to payment timeout.

Document B:

Error code **0xA93** occurs when authentication fails.

**Dense Retrieval
(embeddings)**

Rank **A** high → semantic match

Lexical retrieval (BM25)

Rank **B** high → exact match:
0xA93

Combine both!

Compute two scores:

$$score_{final} = \alpha \cdot score_{dense} + \beta \cdot score_{BM25}$$

OR

Step 1:

Retrieve Top-20 from dense search.



Step 2:

Retrieve Top-20 from BM25.



Step 3:

Merge and deduplicate.



Reranking!

Retrieval- Hybrid search

Dense retrieval struggles with:

- Rare tokens
- Product names
- IDs
- Legal references

Lexical retrieval struggles with:

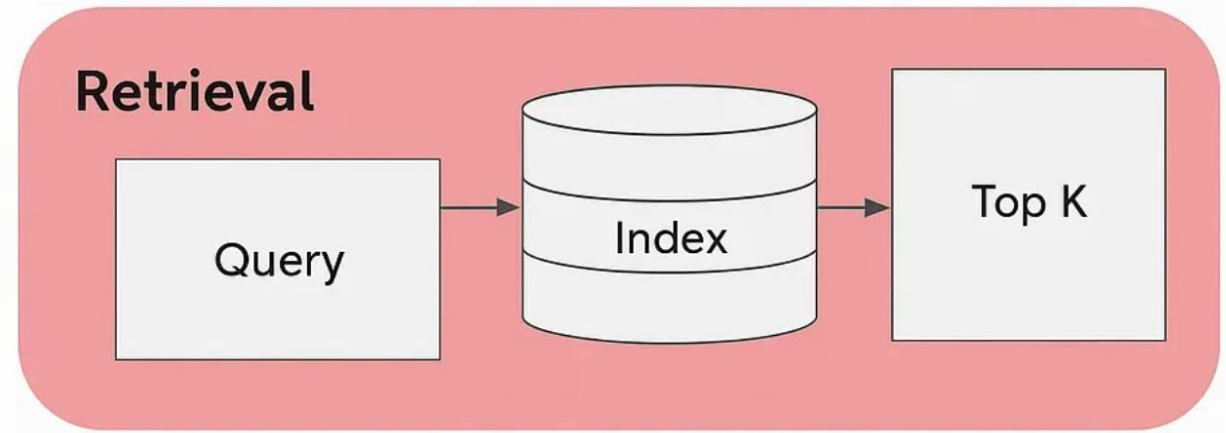
- Paraphrases
- Natural questions

Hybrid = bias–variance trade off.

No single famous paper that “proposed” hybrid retrieval.

Hybrid search is an engineering evolution → used by **search engines** (google, Bing) existed **before RAG!**

Retriever setting



- **Top-k**: start **5** (raise to **10** if misses; lower if noisy/costly)
- **Threshold (similarity)**: start **0.78** (raise 0.82 = cleaner; lower 0.74 = fewer empty retrieves)
- **Filters**: where to focus? (**region/version/department**) to avoid wrong doc → department="Finance" AND region="UK" AND version_date >= 2025-01-01
- **Hybrid**: **0.7 vector / 0.3 keyword** (shift toward keyword for IDs/codes)

Evaluation

- Retrieval quality
- Answer correctness
- Faithfulness

And more

Next steps

- RAG Evaluation
- RAG Variants
- RAG Vs Fine-tuning

Paper this week!

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Additional Material:

- What is Reranking? <https://towardsdatascience.com/rag-explained-reranking-for-better-answers/>
- BM25 model <https://www.elastic.co/blog/practical-bm25-part-2-the-bm25-algorithm-and-its-variables>
- **ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT**
- Dense Passage Retrieval for Open-Domain Question Answering

Thank you!

If you want to ask us any questions about the module, the university, or anything else, drop us an email.

Pro tip: write **COMM117** on the email subject line, to make sure we give it priority.



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Questions?